Enhancing Metabolite-Disease Association Prediction via Matrix Completion and graph neural networks

Pengli Lu*, Jian Zhang

Abstract—**As products of chemical reactions occurring within cells, metabolites are involved in key processes such as cell growth, differentiation, functional maintenance, and immune responses, and they play a significant role in sustaining human life, body growth, and reproduction. Studies have shown that the concentrations of metabolites in patients differ from those in healthy individuals, and leveraging metabolites to predict diseases is crucial for disease diagnosis and therapeutic strategies. However, traditional biological experimental methods are time-consuming and costly, and current computational methods have limitations in terms of accuracy and interpretability. In light of these constraints, we propose a novel deep learning model named AGKformer. This model first utilizes the ADM algorithm to solve the minimum nuclear norm to complete the original sparse matrix, improving the sparsity of the data and providing a richer set of features for subsequent neural network learning. Subsequently, AGKformer extracts domain-specific features from nodes using Graph Convolutional Networks (GCN) and employs a multi-head attention mechanism to capture long-range dependencies, thereby compensating for the shortcomings of GCN. Additionally, we introduce FastKAN for the first time in disease prediction, which enhances the model's interpretability compared to traditional neural networks. Through five-fold cross-validation, AGKformer achieved average AUC and AUPR values of 97.32% and 97.34%, respectively, outperforming existing methods and demonstrating its effectiveness and accuracy in predicting disease-related metabolites. We also conducted ablation experiments to verify the contribution of each module within the model and its overall robustness. Finally, through case studies, we further demonstrate that AGKformer is a reliable tool in discovering potential metabolites.**

Index Terms—**Disease-metabolite associations, Matrix Completion, Graph convolutional networks, Multi-Head Attention, FastKAN**

I. Introduction

Metabolites have a complex and multidimensional relationship with diseases, playing a crucial role in maintaining the stability and health of the body's internal environment [1]. Metabolites are the products of chemical reactions that occur within cells and are involved in key processes such as cell growth, differentiation, functional maintenance, and immune responses. Imbalances or disorders in metabolic regulation are associated with a variety of diseases, including diabetes [2], obesity [3], cardiovascular [4] diseases, and fatty liver diseases. For instance, metabolic abnormalities are considered the root cause of the connection between diabetes and liver diseases. Insulin resistance [5], diabetes, obesity, and fatty liver are closely linked to liver fibrosis, cirrhosis, and liver cancer [6], [7].

In disease research, metabolomics helps to reveal the molecular mechanisms between metabolite levels and disease risk [8]. Studies have found that genetic variations that regulate metabolite levels may affect gene expression and disease risk. Metabolic diseases are conditions caused [9] by genetic mutations or environmental disruptions affecting major digestive organs or organs with significant metabolic functions, such as the liver and pancreas. Disruptions in metabolic pathways can lead to a variety of metabolic diseases, including acid-base imbalances, metabolic brain diseases [10], and lipid metabolic disorders [11]. Major metabolic diseases, such as diabetes, obesity, metabolic syndrome, cardiovascular diseases, and inborn errors of metabolism, are all related to metabolic pathway disorders. The role of metabolism is also an increasingly important aspect in cancer research. Cancer cells rely on inefficient fermentative metabolic pathways, a phenomenon known as the "Warburg effect" [12] which is a hallmark of many cancers and reflects the metabolic adaptations that promote cancer cell growth and survival. Cancer cells require their nutrient inputs as the basic components of the cell, shifting from catabolic pathways to anabolic pathways to support their high proliferation rates [13]. Therefore, there is a close connection between metabolites and diseases, and a deep understanding of these connections is of great significance for the prevention, diagnosis, and treatment of diseases.

Traditional biological experiments can propose hypotheses and test the relationship between metabolites and diseases, but these experiments are often time-consuming and labor-intensive. With the advancement of metabolomics and data mining techniques, multiple databases dedicated to studying metabolomics have been established, including METLIN [14], HMDB [15], and gutMGene [16]. Along with this, there are methods for computationally identifying the relationship between metabolites and diseases, which can avoid the shortcomings of traditional experiments. The development of machine learning has given rise to approaches that primarily utilize optimization and complex network algorithms. In 2018, Hu et al. used random walk to identify diseaserelated metabolites [17]. Later, Lei et al. developed a calculation method based

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on KATZ to predict the metabolite–disease associations [18]. This is also the first application of KATZ algorithm in the field of metabolomics. In 2020, Lei et al. proposed a linear neighborhood similarity with improved bipartite network projection algorithm to predict associations between metabolites and diseases [19]. Furthermore, Zhao et al. introduced a Deep-DRM model in 2021 [20], employing graph convolution networks and principal component analysis (PCA) for identifying metabolite–disease associations. In the same year, Zhang et al.proposed LGBMMDA [21], a method that extracts features from various measurements, applies PCA for noise reduction and utilizes the LGBM classifier for analysis. In 2022, Tie et al. proposed a metabolite–disease association prediction algorithm based on DeepWalk and random forest [22]. During the same period, Sun et al.came up with a graph neural network with attention mechanisms to predict the associations between metabolites and diseases [23].In 2023, Gao et al. proposed a method for predicting metabolite-disease associations based on autoencoders and non-negative matrix factorization [24]. Despite the achievements of current predictive models, there are still some limitations.

Compared to general computational methods, the development of deep learning has greatly improved computational efficiency. However, the core idea of existing deep learning models is to propagate and aggregate information through adjacency matrices to learn complex representations of graphs. Apparently, current deep learning methods rely too much on the metabolite-disease association network, and the constructed association network is too sparse, with only 0.77% of the associations being known. This may lead to problems such as information loss, decreased accuracy, and difficulty in capturing complex patterns, which makes the exploration of node information not deep enough and limits the performance of the model. With in-depth research on the GCN architecture [25], its drawbacks have become increasingly apparent. Although it performs well in processing graph-structured data, traditional GCN models may encounter problems such as gradient vanishing and over-smoothing, which make it difficult for the model to distinguish different nodes and limit the model's ability to capture long-range dependencies, thus affecting performance. In addition, research and application of Transformers [26] are rapidly developing, and in recent years, they have achieved performance comparable to or better than the most advanced GNN models in graph-related tasks. Compared to traditional GNNs, they can handle long-range dependencies and complex topological structures in graphs. However, due to high computational complexity, poor generalization ability, and low interpretability, they still require more optimization and improvement in practical applications. This will drive our work forward.

Inspired by the aforementioned issues, we propose the AGKformer, which employs the Alternating Direction Method of Multipliers (ADMM) optimization algorithm to minimize the nuclear norm for matrix completion, and then uses a model based on Graph Convolutional Networks (GCNs) and a Transformer model that incorporates fastKAN to infer the relevance between metabolites and diseases. Firstly, we construct a heterogeneous network based on known association information, an integrated similarity network for metabolites, and an integrated similarity network for diseases. Next, we use the ADMM [27] optimization algorithm to minimize the nuclear norm on theheterogeneous network to complete the association matrix, which will supplement the sparse associations between diseases and metabolites, enhancing accuracy. Then, we use the graph neural network we have built to learn feature embeddings from the input graph structural data. We find that the message passing mechanism of GNN helps to integrate representations and graph structures. The core of the Transformer is the multi-head self-attention mechanism, which allows the model to directly establish connections between any two positions in the sequence, enabling the model to effectively capture long-range dependencies. Combining them will capture more topological information of the graph, and we use fastKAN in place of the Transformer's feedforward network, enhancing its accuracy and interpretability. Finally, a bilinear decoder is used to reconstruct the association matrix between diseases and metabolites. The following summarizes the three main contributions.

- For the first time, we use the method based on the ADMM algorithm to minimize the nuclear norm to complete the sparse disease-metabolite association network, providing more features for subsequent neural network learning.
- We have enhanced the overall accuracy and interpretability of the model by introducing the more explanatory FastKAN to update the original Transformer's feedforward network.
- We propose a novel weighted graph encoder, AGKformer, which inherits the advantages of traditional GNN and the improved Transformer, enabling more comprehensive and in-depth mining of feature information.

We evaluated the performance of AGKformer in a 5-fold cross-validation (5-CV) experiment, and the results outperformed five previous computational methods. Additionally, we conducted case studies based on AGKformer, and the top 10 predicted metabolite-disease pairs can be verified in relevant literature. These results indicate that AGKformer is an effective and feasible model for predicting potential metabolite-disease associations.

II. Preliminaries

In this section, the concepts of databases, metabolite-disease associations, similarity measures, are mainly introduced, which will be utilized in the subsequent work.

A. *Dataset*

HMDB (Human Metabolome Database, https://hmdb.ca/) [15] is a widely used biochemical database that focuses on storing information related to human metabolism. HMDB provides detailed data on small molecules (metabolites) in the human body, which is crucial for studying metabolic pathways in human health and disease. The current version of the database is 5.0. The dataset we extracted from HMDB includes disease IDs (DOID) and related metabolite information. After removing noise from the raw data, a curated set of 657 diseases, 2,624 metabolites, and 7,772 detected and quantified associations was filtered out. Additionally, diseases and metabolites are uniformly named using MeSH Unique ID (Medical Subject Headings) [28] and HMDB ID, respectively, ultimately obtaining 265 diseases, 2,315 metabolites, and 4,763 associations.

B. *Metabolite–disease associations*

In order to delve into the interplay between diseases and metabolites, we constructed an adjacency matrix $A = (a_{i,j})_{N_m * N_d}$ to quantify their relationships. In this matrix, N_m and N_d represent the number of metabolites and diseases, respectively. Specifically, if there is a known scientific association between disease i and metabolite j, then the element a_{ij} in matrix A is marked as 1, conversely, if there is no direct connection between the two, a_{ij} is 0.

In our study, a total of 613,475 potential associations between 265 different diseases and 2,315 metabolites were identified. Among these associations, 4,763 have been scientifically verified as positive samples. To balance the number of positive and negative samples, thereby reducing the impact of sample imbalance on model performance, we randomly selected 4,763 out of the 608,712 unconfirmed associations as negative samples. Ultimately, we obtained a dataset containing 9,526 samples, which will be used for subsequent model training and testing. Through this representation of the adjacency matrix, we can more intuitively understand the correlation between metabolites and diseases, providing a powerful analytical tool for the discovery of biomarkers for diseases and the potential application of metabolites.

C. *Similarity measures*

*1) **Disease semantic similarity**:* Disease semantic similarity [29] can be calculated based on disease information from the MeSH database. The relationships between diseases can be represented by a directed acyclic graph (DAG), based on the topological structure of the disease ontology and the hierarchical relationships between diseases, an information content contribution factor is introduced. The DAG is represented using the tree numbers in the MeSH descriptors as DAG(d) = (d, T(d), E(d)), where $T(d)$ denotes d itself and all the ancestor nodes of d, and $E(d)$ represents the set of directed edges from the ancestor nodes to the disease nodes. The semantic value of the disease $DV(d)$ can be defined as

$$DV(d) = \sum_{n \in T(d)} D_d(n) \tag{1}$$

and the semantic contribution value of disease n to disease d, $D_d(n)$, is defined as follows:

$$D_d(t) = \begin{cases} 1, & n = d \\ \max\left\{\Delta \cdot D_d\left(n'\right) \mid n' \in \text{children of } d\right\}, & n \neq d \end{cases} \tag{2}$$

where Δ is the semantic contribution decay factor, generally defined as 0.5, which affects $D_d(n)$ between the parent node d and its child node n'. $D_d(n)$ is inversely proportional to the distance between diseases d and n.

Generally speaking, a greater number of shared ancestors in two disease DAGs indicates a higher similarity between them. Therefore, the semantic similarity of diseases d_i and d_j is defined below:

$$DSS(d_i, d_j) = \frac{\sum_{t \in T(i) \cap T(j)} \left(D_i(t) + D_j(t)\right)}{DV(i) + DV(j)} \tag{3}$$

*2) **Gaussian kernel similarity for disease and metabolite**:* The Gaussian kernel [30] is based on the Gaussian (normal) distribution and is used to measure the similarity or distance between data points. Using known topological information from the disease-metabolite association network, kernel methods are employed to construct a kernel function from feature vectors in order to extract associated features. The Gaussian kernel similarity network [31] of diseases and metabolites can be calculated as follows:

$$\mathrm{DGS}(d_i, d_j) = \exp\left(-\omega_d \left\|\mathrm{IP}(d_i) - \mathrm{IP}(d_j)\right\|^2\right) \tag{4}$$

$$\mathrm{MGS}(m_i, m_j) = \exp\left(-\omega_m \left\|\mathrm{IP}(m_i) - \mathrm{IP}(m_j)\right\|^2\right) \tag{5}$$

where IP (d_i) and IP(d_j) represent the feature mappings of the data points d_i and d_j,In Equation (4)(5), ω_d' and ω_m' is a weight parameter that regularizes the influence of distance in the similarity measure. here, we set its value as 1, and ω_d and ω_m is computed as Equation (6)(7):

$$\omega_d = \omega_d' / \left(\frac{1}{n_d} \sum_{i=1}^{n_d} \mathrm{IP}(d_i)^2\right) \tag{6}$$

$$\omega_m = \omega_m' / \left(\frac{1}{n_m} \sum_{i=1}^{n_m} \mathrm{IP}(m_i)^2\right) \tag{7}$$

*3) **Metabolite molecular fingerprint similarity**:* Metabolite molecular fingerprint similarity [24] refers to the assessment of structural similarity between different metabolite molecules by comparing their fingerprints in the context of metabolomics research. A molecular fingerprint is an encoding method that represents the molecular structure; the more similar the molecular fingerprints of two metabolites, the higher their similarity score. The chemical structure of the metabolites, extracted in the standard SMILES format, is converted into a set of binary molecular fingerprints. Thereafter, the Tanimoto coefficient [32], [33] is utilized to quantify the differences between various metabolites. The specific formula is as follows:

$$MFS(m_i, m_j) = \frac{c}{a + b - c} \tag{8}$$

where a and b represent the number of occurrences of 1 in the molecular fingerprints of metabolites m_i and m_j, respectively, and c indicates the number of common occurrences of 1 in both metabolite m_i and m_j.

III. METHDOS

Figure.1 illustrates the workflow of AGKformer. This model proposed in this paper is divided into three to four steps. First, we construct the disease similarity matrix NSD using disease semantic similarity combined with the Gaussian kernel similarity of diseases, and the metabolite similarity matrix MSN

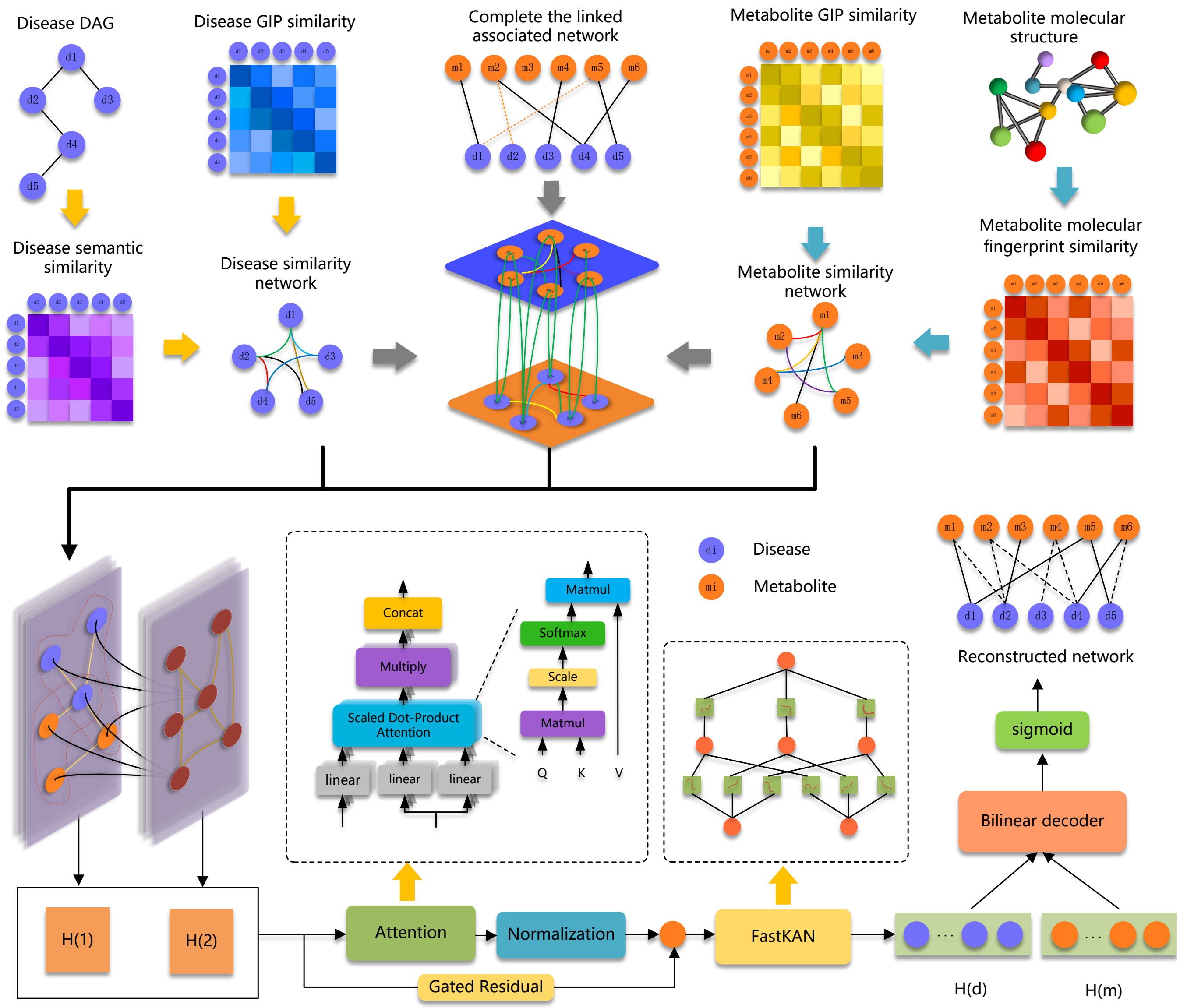

Fig. 1. Overall flow chart of the AGKformer mode.

using the Gaussian kernel similarity of metabolites combined with the similarity of metabolite molecular fingerprint graphs. Based on the ADMM optimization method that minimizes the nuclear norm, we complete the known associations between metabolites and diseases. Then, we construct a heterogeneous network using the similarity matrices DSN and MSN, as well as the completed association matrix. Subsequently, we use this heterogeneous network as input to the GCN, where the GCN encoder is employed for domain feature learning. It is then fed into our improved Transformer encoder, which consists of Multi-Head Attention and fastKAN, and where we have utilized residual connections to enhance training efficiency. This encoder is more effective at modeling long-distance dependencies between nodes in the graph and can learn node representations from multiple subspaces. Finally, a bilinear decoder is used to decode the obtained metabolite-disease embeddings in order to reconstruct the metabolite-disease adjacency matrix and infer potential metabolite-disease relationships.

A. *Construct disease and metabolite similarity networks*

To address the issue of some diseases not having corresponding MeSH molecular descriptions, which leads to a semantic similarity of zero, we will integrate the Gaussian kernel similarity of diseases to construct a comprehensive similarity matrix DSN. This approach helps to mitigate the situation by providing an alternative measure of similarity that does not rely solely on semantic descriptions, thereby enhancing the robustness of the similarity assessment.

$$\mathrm{DSN}(d_i,d_j)=\begin{cases}\mathrm{DGS}(d_i,d_j), & \mathrm{DSS}(d_i,d_j)=0\\ \frac{\mathrm{DGS}(d_i,d_j)+\mathrm{DSS}(d_i,d_j)}{2}, & \text{otherwise}\end{cases} \tag{9}$$

Similarly, for metabolites, there are cases where metabolites do not have corresponding molecular fingerprint profiles, resulting in a molecular fingerprint similarity of zero. To address this, we combine the Gaussian kernel similarity of metabolites to construct a comprehensive similarity matrix MSN.

$$\mathrm{MSN}(m_i, m_j) = \begin{cases} \mathrm{MGS}(m_i, m_j), & \mathrm{MFS}(m_i, m_j) = 0 \\ \frac{\mathrm{MGS}(m_i, m_j) + \mathrm{MFS}(m_i, m_j)}{2}, & \text{otherwise} \end{cases} \tag{10}$$

As we know, there is no association between disease-disease and metabolite-metabolite, therefore we construct a comprehensive similarity between diseases and metabolites to complement the correlation among nodes of the same type, to better mine the information in the graph, and to better predict potential links. In addition, it is also very important for understanding biological processes and the pathogenesis of diseases.

B. *Matrix Completion Algorithm*

After data processing, our final dataset includes 2,315 metabolites and 265 diseases, with a total of 613,475 association pairs, of which only 4,764 are known associations. This indicates that the association network is quite sparse, with known associations accounting for only 0.77%.The original association matrix only contains elements 0 and 1, indicating whether there is an association, and is relatively sparse, making the prediction of unknown associations relatively difficult. Furthermore, we recognize that functionally similar metabolites may be associated with functionally similar diseases. Therefore, by analyzing the functional similarities between metabolites and diseases, we propose a method based on the ADMM optimization algorithm to minimize the nuclear norm to discover their potential connections, thereby replacing element 0, providing a complete matrix for subsequent predictions, and enhancing the accuracy and robustness of model predictions. The process is shown in Figure.2.

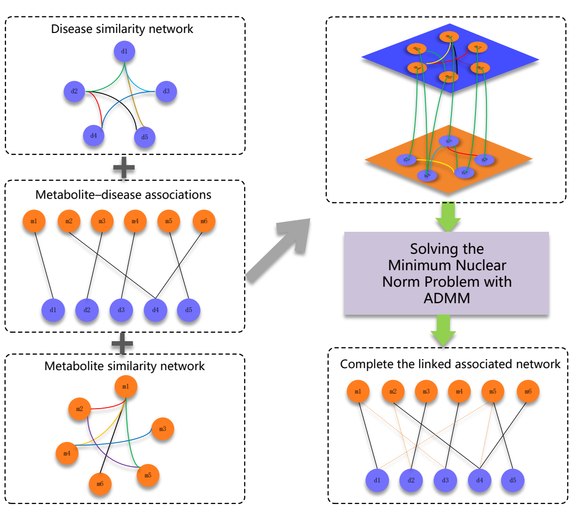

Fig. 2. Overview of Methods for Matrix Completion.

*1) **Construct the target matrix**:* To avoid problems caused by the sparsity of the matrix, such as a decrease in prediction accuracy, this paper solves the minimization of the nuclear norm to complete the matrix. First, based on the previously constructed metabolite and disease similarity networks and the metabolite-disease interaction adjacency matrix, a heterogeneous metabolite-disease network T is built as our target matrix, represented as follows:

$$T = \begin{bmatrix} MSN & A \\ A^T & DSN \end{bmatrix} \tag{11}$$

It is not difficult to deduce that the target matrix T is of dimension $(n+m) \times (n+m)$, that is, 2315×2315 in size. The purpose of constructing the target matrix T is to subsequently use the predicted scores of metabolite-disease interactions obtained from the solution to fill in the missing values.

*2) **Complete the matrix**:* Since the target matrix is a low-rank matrix, we transform the objective into minimizing the rank of the target matrix [34], which can also be a problem of minimizing the matrix's nuclear norm [35]. During this process, we found that the ADMM algorithm [27] not only has fast decomposition speed but also achieves good performance in terms of "slimming down" the matrix. Therefore, we ultimately transform the problem into:

$$\begin{aligned} & min_X \|X\|_* \\ \text{s.t.} \quad & P_\Omega(X) = P_\Omega(T) \end{aligned} \tag{12}$$

where $\|X\|_*$ represents the nuclear norm of X, Ω is the set of coordinates of the target matrix nodes, corresponding to the known metabolite-disease pairs. P_Ω is the orthogonal projection operator on Ω.

$$(P_\Omega(X))_{ij} = \begin{cases} X_{ij}, & (i,j) \in \Omega \\ 0, & \text{else} \end{cases} \tag{13}$$

In addition to this, in order to further enhance the ADMM algorithm, we have included a regularization term and matrix value constraints in the equation to ensure that the predicted scores fall within the range of (0, 1). Furthermore, we have introduced an auxiliary matrix W in the function to further improve the model's convergence performance. The key objective equation is obtained as follows:

$$\begin{aligned} & min_x \|X\| + \frac{\alpha}{2} \|P_\Omega(W) - P_\Omega(T)\|_F^2 \\ \text{s.t.} \quad & X = W, \quad 0 < W_{ij} < 1 \quad (0 \le i, j \le n+m) \end{aligned} \tag{14}$$

where α is the parameter characterizing the error term, $0 \le X_{ij} \le 1$ represents that all elements in X are within the range (0, 1). W is the auxiliary matrix. On the basis of the aforementioned formula, the Augmented Lagrangian function is:

$$\begin{aligned} L(W, X, Y, \alpha, \beta) = \|X\|_* + \frac{\alpha}{2} \|P_\Omega(W) - P_\Omega(T)\|_F^2 + \\ \mathrm{Tr}(Y^T(X - W)) + \frac{\beta}{2} \|X - W\|_F^2 \end{aligned} \tag{15}$$

where Y is the Lagrange multiplier, $\beta > 0$ represents the penalty parameter. We initialize W, X, and Y to $P_\Omega(T)$, and then perform iterations. At the k-th iteration, according to ADMM, we can compute W_{k+1}, X_{k+1}, and Y_{k+1}. When the iteration terminates, the final prediction matrix W^* can be divided according to the previous form of T, where

A^* represents the completed metabolite-disease association matrix, and at this point, the null values in A^* are filled with potential association scores. The prediction matrix and the termination condition for the iteration are expressed as follows:

$$\begin{gathered}W^* = \begin{bmatrix} MSN^* & A^* \\ A^{*T} & DSN^* \end{bmatrix} \\ d1_{k+1} = \frac{\|X_{k+1} - X_k\|_F}{\|X_k\|_F} \leq \text{tol}_1, \\ d2_{k+1} = \frac{|d1_{k+1} - d1_k|}{\max\{|d1_k|, 1\}} \leq \text{tol}_2\end{gathered} \tag{16}$$

C. Graph Encoder

This section will provide a detailed introduction to the graph encoder we have designed. Through collaborative learning with GCN and our improved Transformer, it comprehensively mines the feature information of the graph, enhances the overall accuracy and interpretability of the model, and achieves more efficient encoding of neighborhood information along different paths.

*1) **GCNLayer**:* GCN [25] is a type of neural network designed for processing graph-structured data. The core idea of GCN is to aggregate the feature information of each node's neighboring nodes as well as its own features, and then learn from them through a neural network. First, the adjacency matrix G of the graph is standardized to obtain $\tilde{G}$, which is typically achieved by multiplying the inverse of the degree matrix D with its square root, that is:

$$\tilde{G} = D^{-\frac{1}{2}} G D^{-\frac{1}{2}}. \tag{17}$$

Next, the standardized adjacency matrix $\tilde{G}$ is multiplied with the node feature matrix X to achieve the aggregation of feature information.Finally, the learned weight matrix W is multiplied with the aggregated features through matrix multiplication to obtain the node feature representation for the next layer.The graph convolution operation of GCN can be represented by the following formula:

$$H = D^{-\frac{1}{2}} G D^{-\frac{1}{2}} X W \tag{18}$$

where D is the degree matrix of the input graph, X is the embedding of the nodes at layer L, and W is the layer-specific trainable weight matrix.To better utilize the similarity characteristics, we set the input adjacency matrix to be:

$$G = \begin{bmatrix} MSN & A^* \\ A^{*T} & DSN \end{bmatrix} \tag{19}$$

G is constructed from the integrated similarity matrices MSN and DSN of metabolites and diseases, as well as the completed adjacency matrix A^* we described earlier.We then initialize the feature matrix as:

$$X = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \tag{20}$$

Finally, according to the formula, we learn from the input graph to obtain a new feature matrix as subsequent input.

*2) **Multi-Head Attention**:* Here, we have implemented the multi-head attention [36] mechanism using a graph transformer [37], serving as a message-passing neural network layer capable of learning and capturing long-range dependencies. Specifically, for the node features $H \in \mathbb{R}^{n \times d}$ after graph convolution,We use learnable weights to compute the queries, keys, and values corresponding to each node feature h_i, respectively:

$$Q_i = W_q h_i, \; K_i = W_k h_i, \; V_i = W_v h_i, \tag{21}$$

then divide the transformed results into multiple heads. For each head, we calculate the dot product of the query q_i and all keys k_j, then divide by a scaling factor (usually the square root of the output dimension) to obtain the unnormalized attention scores. Then, normalize the attention scores using the softmax function [38] to obtain the attention weights:

$$\alpha_{i,j} = softmax(\frac{Q_i^\top K_j}{\sqrt{d_k}}) \tag{22}$$

where d_k is the dimension of the key vector.

Multiply the normalized attention weights by their corresponding values v_j, and sum over all nodes to obtain the aggregated features:

$$x_i = \sum_{j \in N_{(i)}} \alpha_{i,j} V_j \tag{23}$$

where $N_{(i)}$ denotes the set of neighboring nodes of node i, then, we use concatenation to join the outputs of all heads:

$$m_i = \text{Concat}(x_i^1, x_i^2, ..., x_i^H) \tag{24}$$

Additionally, we employ residual connections [39] to combine the original node features with the aggregated features, addressing the issue of vanishing gradients:

$$h_i' = W_o h_i + m_i \tag{25}$$

where W_o is the learnable weight matrix. In the end, the output feature h_i' of node i is a combination of the aggregated features and the skip connection. In summary, based on the above formulas, for the GCN output features H, the multi-head attention captures deeper embedding information, and the final output features are:

$$x_i' = W_o x_i + \sum_{j \in N_{(i)}} \text{softmax}\left(\frac{(W_q x_i)^\top (W_k x_j)}{\sqrt{d_k}}\right) W_v x_j \tag{26}$$

*3) **fastKAN**:* To enhance nonlinearity and increase the model's expressive power, as well as to help the model understand the global and local structures in the data, we introduce fastKAN [40], which offers superior accuracy and interpretability compared to traditional Feedforward Neural Networks.

Inspired by the Kolmogorov-Arnold representation theorem, a new type of neural network architecture called Kolmogorov-Arnold Networks (KAN) [41] was born. Unlike traditional Multi-Layer Perceptrons (MLP) [42], the core idea of KAN is to replace each weight parameter with a learnable univariate function, usually a parameterized spline function, thus the entire network contains no linear weights. Specifically, KAN represents the function $f(x)$ as:

$$f(\mathbf{x}) = \sum_q \Phi_q \left(\sum_p \phi_{q,p}(x_p) \right) \tag{27}$$

where Φ and $\Phi_{q,p}$ are univariate functions. The design of KAN allows it to break down complex functions into simpler components that are easier for the network to grasp. The network uses B-splines to construct a family of functions that can effectively approximate smooth univariate functions. However, during the training process, if variables exceed the set domain, adjustments to the spline grid are necessary. This adjustment process requires recalculating the B-spline [43] basis through deBoor-Cox iteration, which may lead to efficiency bottlenecks in KAN.

To improve efficiency while maintaining accuracy, we have adopted FastKAN as a replacement for traditional methods. FastKAN uses Gaussian kernel radial basis functions (RBFs) to approximate the third-order B-spline basis. Furthermore, layer normalization processing is applied to prevent input data from exceeding the range of the RBFs.

The basic idea of Radial Basis Function (RBF) [44] is to model a function by combining several radially symmetric functions, each centered at different points in the input space. The output of an RBF network is a linear combination of these radial basis functions, weighted by adjustable coefficients:

$$f(x) = \sum_{i=1}^{N} w_i \phi\left(\|x - c_i\|\right) \tag{28}$$

where w_i is the learnable weight, which depends on the distance between the input x and the center c_i. The Gaussian function is a common RBF:

$$\phi(\|x - c_i\|) = \exp\left(-\frac{\|x - c_i\|^2}{2h^2} \right) \tag{29}$$

where $\|x - c_i\|$ is the radial distance, and h is a parameter that controls the width or spread of the function.

Specifically, for the features we learned earlier, we apply layer normalization, then generate high-dimensional features through RBF, and next, flatten the high-dimensional features generated by RBF and use the spline linear layer to map them to the output dimensions. Finally, after processing the original input features through an activation function and linearly mapping them, we add them to the output of the RBF features to obtain the output of the generated layer. This will further help us extract feature information.

D. *Decoder*

After encoding the node features through our encoder to capture global and long-range structural information, we obtain the final embeddings for metabolites and diseases. Subsequently, we utilize a bilinear decoder to transform the node embeddings into the association probabilities between metabolites and diseases, resulting in the final association matrix:

$$H^{(2)} = \begin{bmatrix} {H_m}^{(2)} \\ {H_d}^{(2)} \end{bmatrix} \tag{30}$$

$$A' = sigmod\left({H_m}^{(2)} W ({H_d}^{(2)})^T \right) \tag{31}$$

where $H_m^{(2)} \in \mathbb{R}^{2315\times512}$ and $H_d^{(2)} \in \mathbb{R}^{265\times512}$ are the final embeddings of the diseases and metabolites, and $W \in \mathbb{R}^{512\times512}$ is the trainable weight.

Additionally, in classification tasks, the cross-entropy loss function [45] is a very popular choice because it can quantify the difference between the predicted probability distribution and the true label's probability distribution. The specific implementation involves:

$$L = -y\log(\hat{y}) - (1-y)\log(1-\hat{y}) \tag{32}$$

where y = 0 or 1 represents the true label, and $\hat{y} \in [0, 1]$ represents the predicted probability value of the association between the disease and the metabolite.

Meanwhile, the Adam optimizer [46] is used to optimize the minimum value and update parameters, which accelerates the model's convergence. To prevent overfitting, dropout [47] is also used during the encoding process, with the aim of training different subnetworks in different batches.

IV. Result

A. *Evaluation indexes*

After constructing our AGKformer model, we employed evaluation metrics to measure its performance. To thoroughly assess the model's effectiveness and select the optimal configuration, we utilized the efficient and practical machine learning method of five-fold cross-validation [48]. This approach fully leverages the dataset by evaluating the model's performance on different subsets, aiding us in more accurately selecting the best model or adjusting hyperparameters.

Specifically, we evenly divided all samples into five subsets of equal size. In each round of cross-validation, we used four subsets as the training set to train the model, while the remaining subset served as the test set to evaluate its performance. This method ensures that every sample has the opportunity to be used as test data, guaranteeing comprehensiveness and reliability in model assessment. It not only enhances the efficiency of data utilization but also helps us more precisely estimate the model's generalization ability. As the threshold in the model changes, the True Positive Rate (TPR) and False Positive Rate (FPR) can be calculated as follows:

$$TPR = \frac{TP}{TP + FN} \tag{33}$$

$$FPR = \frac{FP}{TN + FP} \tag{34}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive and false negative, respectively.

To visually demonstrate the performance of our proposed model, we plotted the Receiver Operating Characteristic (ROC) curve. In this plot, the x-axis represents the false positive rate, while the y-axis represents the true positive rate. The area under the ROC curve, known as the AUC value, measures the model's ability to discriminate between different classes. To reflect the model's enrichment capability for true samples, we used the relationship between precision and recall

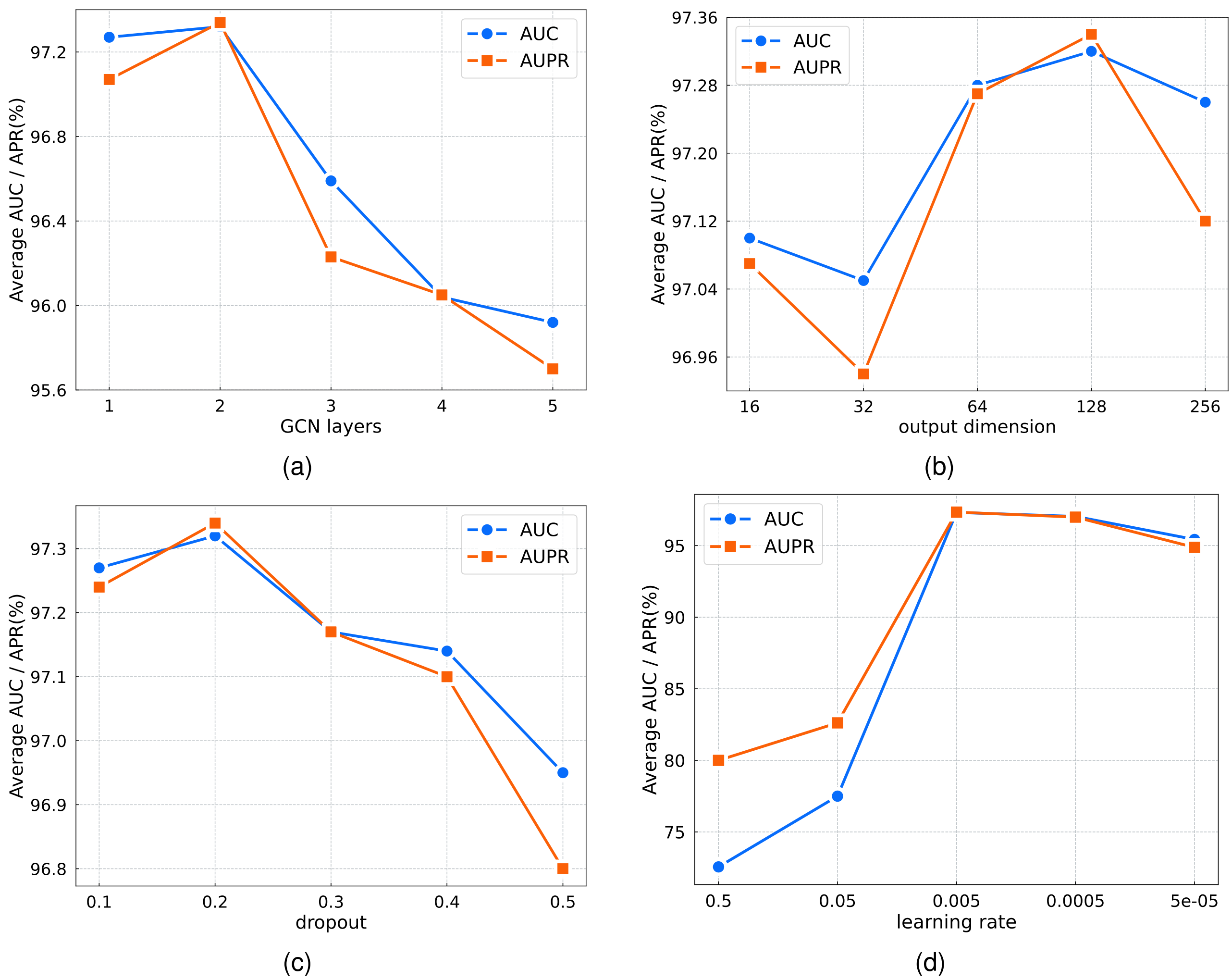

Fig. 3. Effect of parameters in AGKformer model on average AUC and AUPR. (a) GCN layers, (b) output dimension, (c) dropout, (d) learning rate.

to obtain the AUPR value, which is the area under the Precision-Recall curve. Additionally, we employed accuracy (Acc), precision (Pre), recall, and the F1 score as evaluation metrics:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{35}$$

$$Precision = \frac{TP}{TP + FP} \tag{36}$$

$$Recall = \frac{TP}{TP + FN} \tag{37}$$

$$F1 = \frac{1}{\frac{1}{Precidion} + \frac{1}{Recall}} \tag{38}$$

B. Parameter experiment

Following the modeling of the associations between metabolites and diseases using matrix completion and neural networks, we implemented the AGKformer using the PyTorch deep learning framework. During the training process, we fine-tuned the model with numerous parameters, and to expedite model convergence, we also utilized the Adam optimizer. In the five-fold cross-validation, each fold was trained for 200 epochs, with hyperparameters such as the number of GCN layers, the number of multi-head attention heads, dropout rates, and learning rates all impacting the model's performance.

We used the method of holding variables constant to determine the optimal values for each parameter. We selected the number of GCN layers from the set {1, 2, 3, 4}, select the output dimension of the encoder from the set {32, 64, 128, 256}, the learning rate from the set {0.009, 0.007, 0.005, 0.003, 0.001}, and dropout from the set {0.1, 0.2, 0.3, 0.4, 0.5}. By Figure.3 illustrating the impact of these parameters on AUC and AUPR under five-fold cross-validation, we can observe that the algorithm performs optimally and maintains a balance in terms of performance and complexity when there are two GCN layers,The output dimension is 128, a learning rate of 0.005, and a dropout rate of 0.2.

Additionally, we have conducted experiments and filtered other parameters of model. In the calculation of disease semantic similarity, we set Δ to 0.5, and the kernel bandwidth ω' in the Gaussian kernel similarity is set to 1. For matrix completion, we set the penalty parameter ρ of the ADMM algorithm to 1, and the threshold parameter α to 0.01. In five-fold cross-validation, we chose Adam as the optimizer and selected the binary cross-entropy function as the loss function.

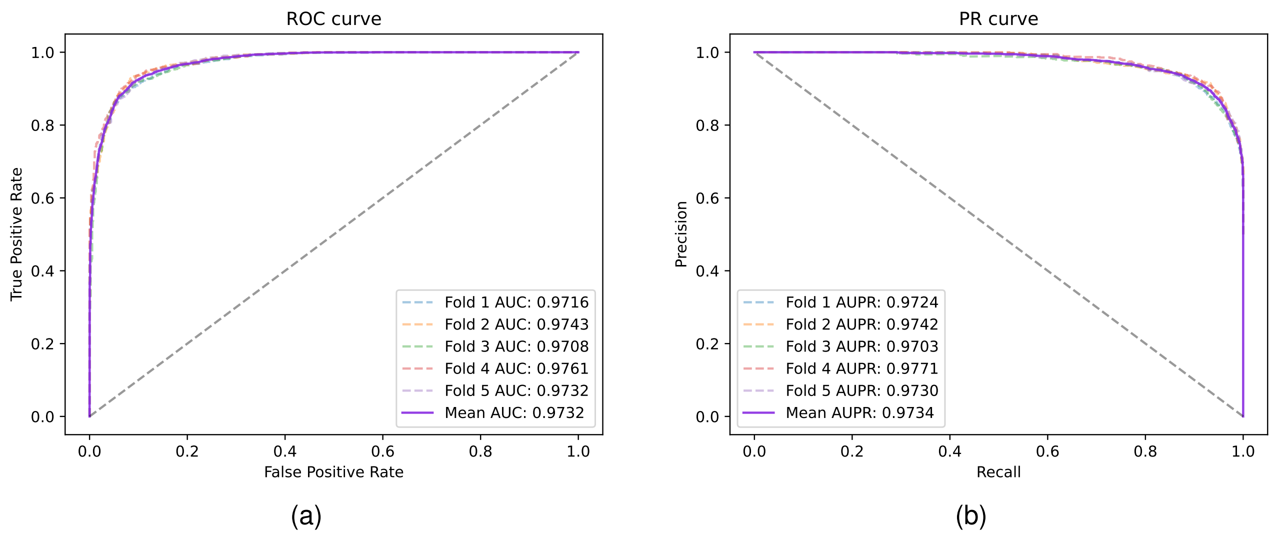

Fig. 4. (a) ROC curves of the AGKformer model under five-fold cross-validation. (b) PR curves of the AGKformer model under five-fold cross-validation.

For the selection of our model parameters, see Table I.

TABLE I
THE PARAMETERS OF AGKFORMER

Structure	**Parameters**	**value**
ADMM	ρ	1.0
	α	0.01
	tol	1e-4
Encode	GCN layers	2
	Output dimension of Layer1	64
	Output dimension of Layer2	64
	The heads of MHA	1
	Output dimension of MHA	64
	dropout	0.2
	learning rate	0.005
	Loss function	binary cross entropy
	Optimizer	Adam
	Epoch	200
	init scale	0.1
	grid min	-2
	grid max	2
	num grids	8
	base activation	silu
Decode	gain	1.414

After parameter tuning, the optimal results were achieved, and in TableII and Figure4, we provide feedback on the comprehensive performance of the model. From Table II, it can be seen that under the optimal parameters, the average Acc, Pre, Recall, and F1 scores of the AGKformer model reached 91.68%, 90.91%, 92.61%, and 91.75%, respectively. In addition, as shown in Figure 4(a), the average AUC of the AGKformer model's five-fold cross-validation reached 97.43%, which is the average of 97.37%, 97.44%, 97.50%, 97.51%, and 97.31%. As shown in Figure 4(B), the average AUPR of the AGKformer model's five-fold cross-validation reached 97.40%, which is the average of 97.24%, 97.30%, 97.55%, 97.58%, and 97.32%. This means that our model has good accuracy, strong generalization ability, and superior interpretability.The outstanding performance of the AGKformer model can be attributed to the following reasons: firstly, we use the ADMM optimization algorithm to solve the heterogeneous matrix with minimum nuclear norm to complete the association matrix, making the original association matrix no longer sparse, which is more conducive to our more precise predictions. Secondly, we combine GCN with the introduced FastKAN multi-head attention to extract features from the heterogeneous network, learning domain features while further learning global features. In addition, the introduction of FastKAN has improved the model's interpretability and also demonstrated the model's good generalization ability. Through training the model and parameter tuning, the best performance is achieved.

TABLE II
THE PARAMETERS OF AGKFORMER

Fold	Acc(%)	Pre(%)	Recall(%)	F1(%)
1	91.02	91.85	90.02	90.93
2	92.33	91.55	93.28	92.40
3	91.07	90.39	91.91	91.15
4	91.96	90.56	93.70	92.10
5	91.20	89.39	93.51	91.40

C. Ablation experiments

To further evaluate the performance and robustness of AGKformer, we performed an ablation study on our model's modules to determine the contribution of each submodule. While keeping other modules unchanged, we removed one module at a time and conducted experiments on the same dataset.

- Del-ADMM: We do not use the completed association matrix; instead, we utilize the original sparse matrix.
- Del-GCN: Remove the GCN layer, leaving only a simplified encoder with multi-head attention and FastKAN.
- Del-MHA: A simplified encoder that retains only the GCN and FastKAN components.
- Del-FastKAN: Remove FastKAN to obtain an encoder composed of GCN layers and multi-head attention.

As shown in Figure.5, the AGKformer model's performance in the ablation study is demonstrated with the impact of

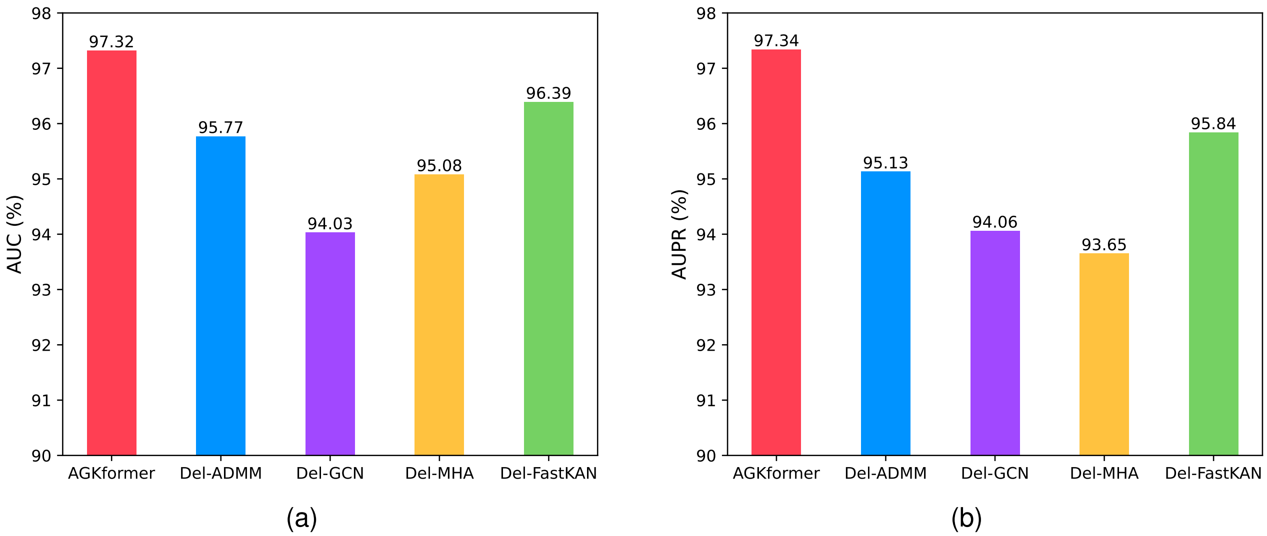

Fig. 5. Average AUPR and AUC of AGKformer and ablation models on five-fold cross-validation

different components on AUC and AUPR. By progressively removing various components from the model, we can assess the contribution of each part to the overall performance. The results indicate that the complete AGKformer model has the highest AUC and AUPR values, reaching 97.32% and 97.34%, respectively. When different components are removed, the model's performance decreases. Del-ADMM's AUC and AUPR values drop to 95.77% and 95.13%, suggesting that the completion of the association matrix can improve model accuracy. Del-GCN further reduces the AUC and AUPR values to 94.03% and 94.06%, highlighting the importance of the GCN layer in feature extraction. Del-MHA's AUC and AUPR values are 95.08% and 93.65%, indicating that the multi-head attention mechanism is beneficial for capturing long-range dependencies. Del-FastKAN's AUC and AUPR values are 96.39% and 95.84%, with a decrease in performance, implying that FastKAN can accelerate model convergence and enhance accuracy.

Overall, the ablation study results emphasize the significance of each component in the AGKformer model, especially the GCN layer and ADMM, which have a notable impact on the model's final performance. Through this analysis, we can better understand the model's internal working mechanisms and provide guidance for future model optimization.

D. Comparison with benchmark approaches

To intuitively demonstrate the advantages of AGKformer, we compared it with the association prediction methods that have emerged in the field of bioinformatics in recent years, and conducted tests and comparative analyses on the same datasets.

- Bi-randomwalks (MDBIRW) [49] predicts disease-related metabolites by reconstructing disease similarity networks and metabolite functional similarity networks, and then performing a double random walk algorithm on these networks.
- DWRF [22] integrates multiple similarity metrics for diseases and metabolites, applies DeepWalk for feature extraction, and then uses random forests to predict the associations between metabolites and diseases.
- Deep-DRM [20] constructs networks of metabolites and diseases, uses GCN to encode features, applies PCA for dimensionality reduction, and then identifies related metabolites through a deep neural network.
- GCNAT [23] constructs a heterogeneous network, encodes features with GCN, integrates embeddings using a graph attention layer, and decodes and scores to predict disease-related metabolites.
- MDA-AENMF [24] integrates various similarity networks and uses autoencoders and non-negative matrix factorization to extract features, merging them into a comprehensive feature vector for each metabolite-disease pair to predict potential associations.

After five-fold cross-validation, AGKformer was compared with the aforementioned algorithms. Figure.6 shows a comparison of the evaluation metrics for the four baseline methods. AGKformer demonstrates a significant advantage across all key metrics. Specifically, AGKformer achieves an accuracy (Acc) of 91.52%, which is 0.85% higher than the second-best model, MDA-AENMF. It also reaches a precision (Pre) of 90.75%, a recall (Recall) as high as 92.48%, and an F1 score of 91.60%. Moreover, as significantly shown in Figure.7, AGKformer's ROC and PR curves are superior to those of the other five baseline methods. AGKformer has the largest area under the ROC curve, with an AUC value of 97.32%, which is 7.76%, 2.21%, 2.09%, 2.11%, and 0.77% higher than MDBIRW, DWRF, DeepDRM, GCNAT, and MDA-AENMF, respectively, indicating that our model performs quite ideally. Additionally, AGKformer also has the largest area under the PR curve, at 97.34%, which is 7.32%, 2.19%, 2.45%, 2.20%, and 1.07% higher than MDBIRW, DWRF, DeepDRM, GCNAT, and MDA-AENMF, respectively. This indicates that AGKformer can maintain high accuracy while better identifying all positive samples, highlighting its exceptional performance and potential value in practical applications.

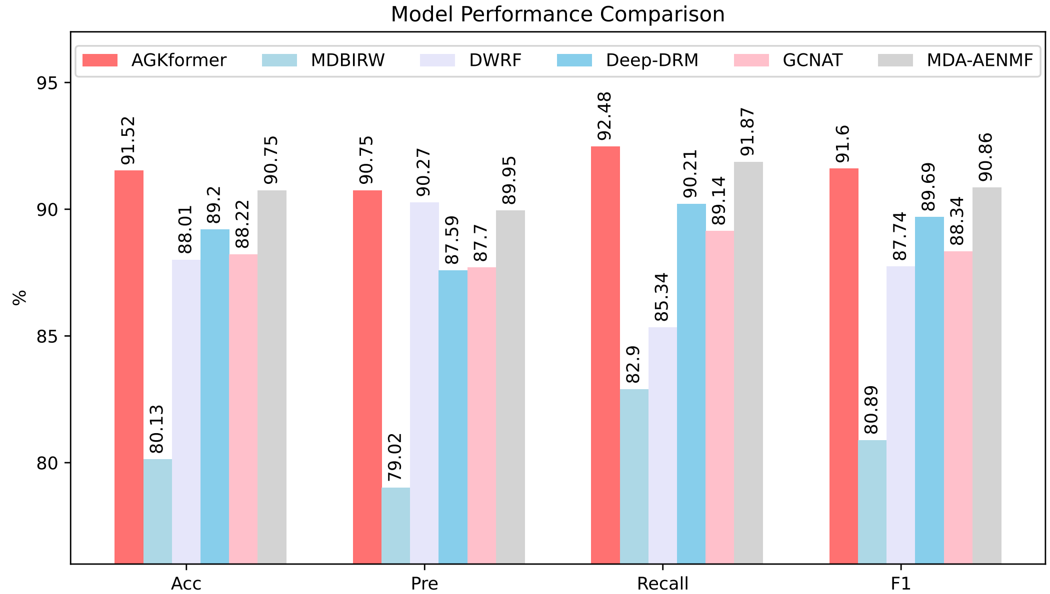

Fig. 6. Performance comparison between the proposed AGKformer model and five benchmark methods based on five-fold cross-validation.

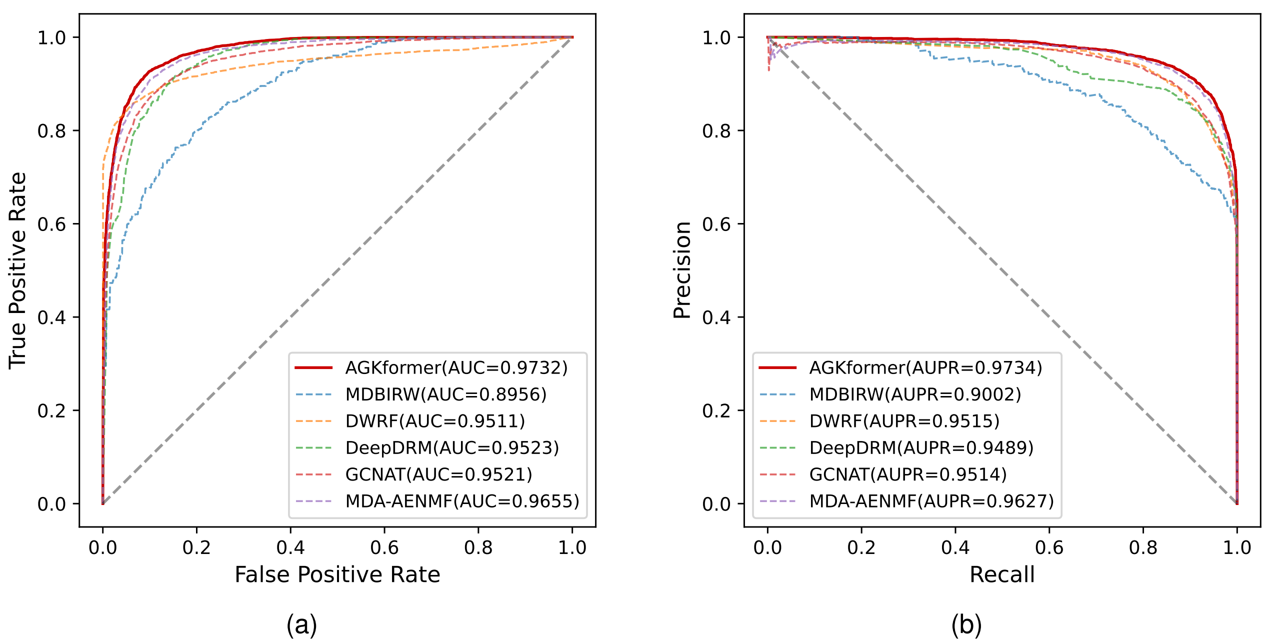

Fig. 7. . (a) ROC curves of the AGKformer model and benchmark methods under five-fold cross-validation. (b) PR curves of the AGKformer model and benchmark methods under five-fold cross-validation.

The aforementioned methods all utilize sparse bipartite heterogeneous networks. Our proposed method is based on similarity networks of metabolites and diseases to complete the metabolite-disease association matrix. This matrix, along with the similarity networks, is then input into a GCN for feature extraction. A multi-head attention mechanism is used to capture long-range dependencies, followed by further nonlinear transformations through a FastKAN feedforward network, and finally, the associations are decoded to obtain the final relationships. This approach not only improves accuracy but also enhances the model's interpretability.

E. Case study

To further evaluate the performance and practical effects of the AGKformer model, we conducted an in-depth analysis of two complex diseases—Alzheimer's disease and Crohn's disease. Specifically, we selected Alzheimer's disease and Crohn's disease along with all associated metabolites as the positive samples for the test set, and randomly sampled an equal number of unknown associations as negative samples. We then used all samples outside of these diseases as the training set. Ultimately, we obtained association scores between specific diseases and metabolites, ranked them, and identified the top 15 associated metabolites. To verify the accuracy of these predictions, we conducted detailed literature searches and data analysis using the HMDB and NCBI databases.

TABLE III
PREDICTED RESULTS OF TOP 15 METABOLITES ASSOCIATED WITH ALZHEIMER'S DISEASE

Rank	Metabolite	Probability	PMID
1	Creatinine	0.9998	23857558
2	alpha-D-Glucose	0.9998	9693263
3	D-Glucose	0.9992	9693263
4	Hypoxanthine	0.9989	23857558
5	Homocysteine	0.9983	21474939
6	Adenine	0.9959	23857558
7	Chloroform	0.9951	None
8	L-Alanine	0.9921	23857558
9	Hexadecanedioic acid	0.9906	None
10	Dopamine	0.9902	17031479
11	Substance P	0.9899	11314776
12	Mannitol	0.9871	8595727
13	L-Arginine	0.9866	17031479
14	L-Proline	0.9855	17031479
15	Choline	0.9850	15465626

TABLE IV
PREDICTED RESULTS OF TOP 15 METABOLITES ASSOCIATED WITH CROHN'S DISEASE

Rank	Metabolite	Probability	PMID
1	L-Serine	0.9629	27609529
2	Glycine	0.9566	25598765
3	Isobutyric acid	0.9506	24811995
4	L-Glutamine	0.9422	27609529
5	D-Aspartic acid	0.9405	17269711
6	D-Glutamic acid	0.9382	17269711
7	L-Alanine	0.9358	17269711
8	L-Tyrosine	0.9340	25598765
9	Hydrocinnamic acid	0.9337	28842642
10	Acetic acid	0.9318	17269711
11	beta-Alanine	0.9293	28842642
12	Creatine	0.9276	27609529
13	L-Lysine	0.9228	17269711
14	L-Threonine	0.9210	17269711
15	Deoxyuridine	0.9186	27609529

Alzheimer's disease is a common neurodegenerative disease that primarily affects the elderly, leading to a gradual decline in memory, thinking, and behavioral abilities [50]. It is the most common form of dementia, and there is currently no cure. Treatment mainly focuses on alleviating symptoms and improving quality of life. The exact cause of Alzheimer's disease is unknown, but it involves a variety of genetic and environmental factors. Currently, some scholars have found that the concentrations of metabolites in Alzheimer's patients differ from those in healthy individuals [51]. They are trying to study and diagnose Alzheimer's disease in advance based on metabolites [52]. With the development of deep learning, some neural network-based methods have emerged in recent years. TableIII lists the top 15 metabolites predicted by our proposed AGKformer model to be highly related to Alzheimer's disease, and 13 of them have been verified. Our model predicts that Creatine [53], alpha-D-Glucose, and D-Glucose [54] are the three most related metabolites to the disease, with probability values of 0.9998, 0.9998, and 0.9992, respectively. Additionally, the verification of other metabolites can be found in the reference literaturesuch as L-Alanine, Hypoxanthine, and L-Phenylalanine. These high probability values indicate a strong association between these metabolites and Alzheimer's disease. In addition, our model also predicts several other metabolites, such as Chloroform and Hexadecanedioic acid, which also show significant relevance to Alzheimer's disease.

Crohn's disease is a chronic inflammatory bowel disease [55] that can affect any part of the digestive tract, but it most commonly affects the terminal ileum and the colon. This disease typically causes chronic inflammation and ulceration of the intestines, leading to a range of discomforts and symptoms. For Crohn's disease, the AGKformer model also performed exceptionally well, predicting that L-Serine [56], Glycine [57], and Isobutyric acid [58] are the three metabolites most closely associated with Crohn's disease, with probability values of 0.9629, 0.9565, and 0.9506, respectively. Additionally, the verification of other metabolites can be found in the reference literature. These high probability values further confirm the potential links between these metabolites and Crohn's disease. The model also predicted other metabolites related to Crohn's disease, such as L-Glutamine, D-Aspartic acid, and D-Glucuronic acid. All of the top 15 predicted metabolites have been verified.

V. CONCLUSIONS

Metabolic disease association research aims to explore the connections between metabolites and diseases to identify biomarkers and therapeutic targets, which is crucial for disease diagnosis, treatment, and prevention. By analyzing changes in metabolites, scientists can uncover disease mechanisms and provide a basis for developing new therapies. Traditional biological experiments are time-consuming and labor-intensive. Although computational methods have emerged in recent years, their accuracy and interpretability need improvement, necessitating new algorithms to address the shortcomings of current approaches. Our proposed AGKformer uses a completed association matrix and deep learning to learn features, achieving more accurate predictions. First, we utilize an ADMM optimization algorithm based on similarity networks of metabolites and diseases to complete the known metabolite-disease association matrix, making the original matrix less sparse. Then, we employ GCN and multi-head attention for feature learning, capturing long-range dependencies more effectively. Finally, by scoring the final embeddings, we obtain the predicted results. Through five-fold cross-validation, compared with baseline methods, AGKformer achieved the best results with AUC(97.32%) and AUPR(97.34%). Additionally, we conducted ablation experiments on our model's modules, verifying the contribution of each module and the overall robustness. Lastly, we conducted case studies on Alzheimer's disease and Crohn's disease, and the results consistently show that our model is precise in predicting disease and metabolite associations.

The performance improvement of the AGKformer model is mainly due to the following factors. First, we use the ADMM algorithm to solve the minimum nuclear norm to complete the original sparse matrix, improving the sparsity of the original data and enhancing the model's predictive accuracy. Second, the rapid development of deep learning has greatly helped improve our model's performance; we use GCN to extract domain features of nodes and learn the underlying patterns of the data. Third, while learning domain features, we use multi-head attention to capture long-distance dependencies, compensating for the shortcomings of GCN. Additionally, we are the first to use FastKAN in disease prediction, which, compared to traditional neural networks, improves the model's interpretability.

DATA AVAILABILITY STATEMENT

The codes and data sets are available online at https://github.com/.

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ETHICS DECLARATIONS

This study was conducted in compliance with the ethical standards and guidelines for research that does not involve human or animal subjects. All authors have reviewed and approved this manuscript for publication.

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